HACKIIITD 2026 · ROUND 1 · OPEN INNOVATION

A voice agent that runs your phone and the web, in your own language

Team

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Members

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Problem: digital services still need someone to operate them

Portals without apps

Scholarship status, ration card, land records, pension and PNR each live on a different website, with English forms, captchas and OTPs.

Language barrier

Many users think and speak in Hindi or a regional language, and cannot navigate English menus or type into complex forms.

Dependence on others

People rely on family, cyber cafés or service centres, paying fees and waiting days for tasks that take a few minutes.

Voice assistants today answer questions. None of them finish the task.

Solution: say it once, the agent does it

1 · SPEAK
Hindi, Hinglish or English, hands-free

2 · UNDERSTAND
On-device model turns speech into an action

3 · ACT
On the phone or on any website

4 · CONFIRM
Asks before risky steps, speaks the result

On the phone · works offline

- Call or message a contact
- Set alarms and reminders
- Read the screen aloud

On the web · via TinyFish

- Check scholarship or pension status
- Look up ration card or land records
- Check PNR and fill government forms

Target users

Elderly users

Comfortable talking, not tapping through menus. Today they wait for family to check pensions and bills.

Rural and low-literacy users

Own a smartphone but struggle with English forms. Voice in their language removes the barrier.

Visually impaired users

Screen readers describe a page; our agent completes the task on it and reads back the result.

Service centres and NGOs

Operators who do the same portal tasks for hundreds of people a day, a B2B channel to scale.

Technical architecture and stack

Wake word
openWakeWord

→

Speech to text
Whisper.cpp / Vosk

→

Intent model
Gemma / Qwen 1–2B

→

Router
Phone task or web task?

ON DEVICE · OFFLINE

Phone executor

Android Intents for calls, SMS, alarms; AccessibilityService to tap, type, scroll and read the screen

CLOUD · TINYFISH

Web agent

Thin FastAPI backend sends goals to TinyFish Agent, Search, Fetch and Vault; results return as JSON

ON DEVICE

Confirm and reply

Voice confirmation before risky actions; Android TextToSpeech reads the result aloud

Stack: Kotlin Android app · MediaPipe LLM Inference / llama.cpp · Python FastAPI · TinyFish API

TinyFish integration: every web task runs through it

TinyFish API	How we use it
Agent	Operates government and utility portals end to end: fills forms, checks status, returns JSON
Search	Finds the right official portal or page for a spoken request
Fetch	Reads static pages like scheme details and notices cheaply, no full agent run
Vault	Stores portal logins; credentials are injected at the browser layer, the AI never sees them

EXAMPLE AGENT GOAL

"Open the scholarship portal, enter application ID [spoken by user], and return the application status as JSON."

Use case: ”Meri scholarship ka status batao”

01

User speaks the request in Hindi

02

Agent detects a web task and asks for the application ID

03

TinyFish opens the portal and fills the form; user speaks the OTP

04

Status returns as structured JSON

05

Agent reads the result aloud in Hindi

No app to download, no English form, no cyber café visit.

Scalability and feasibility

Why it scales

- Speech and intent run on the user's phone, so near-zero server cost per user
- Stateless backend scales horizontally
- A new portal is a new goal template, not a new scraper
- A new language is a speech model swap

Limits we plan for

- Agent cost: cache results, use Fetch for reads
- Captchas and OTPs: the user speaks them in
- Budget phones: smaller model, cloud fallback
- Payments: hand off to the user's own UPI app

FEASIBLE IN 24 HOURS

Android app, 3 phone commands, 2 portal flows via TinyFish

EDGE OVER EXISTING TOOLS

Others answer questions or need flagship phones; we finish Indian portal tasks on any Android

Prototype and finale build plan

PROTOTYPE · KEY SCREENS

LISTENING



"Meri scholarship ka status batao"

CONFIRM

Application ID 2026-XXXX enter karu?

Haan

Nahi

RESULT

Approved

Payment in process

Reading aloud in Hindi

24-HOUR PLAN

Hours 0–8: speech to text and intent pipeline on device

Hours 8–16: phone executor and TinyFish portal flows

Hours 16–24: confirmation layer, polish, live demo

Your phone and the web, operated by your voice.